

Benchmarking TypeSafe's Jev (System One Model) on Biosecurity-Relevant Multiple-Choice Evaluations

An automated evaluation pipeline over WMDP and LAB-Bench

Abstract

We evaluate TypeSafe's Jev (*jev-latest*), a non-generative “System One” classification model, on 6,021 multiple-choice questions spanning the WMDP benchmark (bio, chem, cyber, and a shortcut-robust bio variant) and six text-based subtasks of LAB-Bench. Each item is posed as a single typed *choice* decision against Jev's HTTP API, with the model's returned option and per-option probability distribution scored against ground truth. We find accuracy ranges from 85.2% on WMDP-Bio recall questions down to 36.6% on LAB-Bench SuppQA, and identify two distinct failure regimes: (1) honest low-confidence guessing on tasks requiring multi-step computation or derivation (WMDP-Cyber, LAB-Bench), accompanied by a measurable option-order position bias; and (2) confident, miscalibrated errors on knowledge questions with closely-matched distractors (WMDP-Bio/Chem). Pooled calibration analysis (Expected Calibration Error = 0.091) shows Jev's confidence score is a usable, if imperfect, signal for confidence-gated escalation to a stronger fallback model.

1. Introduction

The Weapons of Mass Destruction Proxy (WMDP) benchmark (Li et al., 2024) measures hazardous knowledge in biosecurity, cybersecurity, and chemical security domains as a proxy for evaluating and unlearning dangerous capabilities in language models. LAB-Bench (Laurent et al., 2024) evaluates practical biology-research capabilities — cloning design, protocol troubleshooting, sequence manipulation, literature and database lookup — that are directly relevant to dual-use biosecurity risk assessment. This report describes a pipeline built to evaluate TypeSafe's Jev model, a proprietary “System One” model that returns typed, probabilistic decisions in a single parallel pass rather than generating free text, against both benchmarks, and analyzes where and why its answers diverge from ground truth.

2. Method

Model interface. Each question is submitted to POST `https://api.typesafe.ai/v1/systemone` with `model=jev-latest`. The question stem and enumerated options (A, B, C, ...) are placed in the `state` field as plain text; a single `choice-type` typed question is attached whose `criteria` map each option letter to its literal text. Jev returns a selected letter, a full probability distribution over letters, and a scalar confidence.

Scoring. The returned letter is mapped back to its original option index and compared against the dataset's ground-truth answer index. Requests are issued concurrently (16 workers) with exponential backoff on HTTP 429/529.

Dataset preparation. WMDP subsets (bio/chem/cyber) and WMDP-Bio-Robust were pulled directly from their published Hugging Face parquet files. WMDP-Bio-Robust encodes its `choices` field as a Python/NumPy array repr mixing single- and double-quoted strings; a quote-aware regex tokenizer was written to parse it losslessly (validated on all 811 rows). LAB-Bench subtasks store answers as an `ideal`

string plus a distractors list rather than a fixed index; options were deduplicated (a small number of rows, notably in SeqQA, contain duplicate distractor text) and shuffled with a fixed seed before submission, with the answer index recorded post-shuffle. Two LAB-Bench subtasks, FigQA and TableQA, encode their supporting evidence as embedded raster images (a photograph/figure or a screenshotted table) rather than text; since Jev's state field accepts only text or structured text/JSON data, these two subtasks were excluded from evaluation rather than scored on the question text alone, which would not reflect a fair or meaningful measurement.

3. Datasets

Dataset	n	Format	Task type
WMDP-Bio	1,273	MCQA (4-opt)	Biosecurity knowledge recall
WMDP-Chem	408	MCQA (4-opt)	Chemical security knowledge recall
WMDP-Cyber	1,987	MCQA (4-opt)	Cybersecurity knowledge + code/arithmetic reasoning
WMDP-Bio-Robust	811	MCQA (4-opt)	Bio knowledge, shortcut-controlled paraphrase set
LAB-Bench CloningScenarios	33	MCQA (var. opt)	Molecular cloning scenario reasoning
LAB-Bench ProtocolQA	108	MCQA (var. opt)	Wet-lab protocol troubleshooting
LAB-Bench SeqQA	600	MCQA (var. opt)	DNA/protein sequence manipulation
LAB-Bench DbQA	520	MCQA (var. opt)	Bioinformatics database lookup
LAB-Bench LitQA2	199	MCQA (var. opt)	Scientific literature fact retrieval
LAB-Bench SuppQA	82	MCQA (var. opt)	Paper supplementary-material retrieval

Table 1. Datasets evaluated. All are 2–9 option multiple-choice; LAB-Bench option counts vary per item (min 2).

4. Results

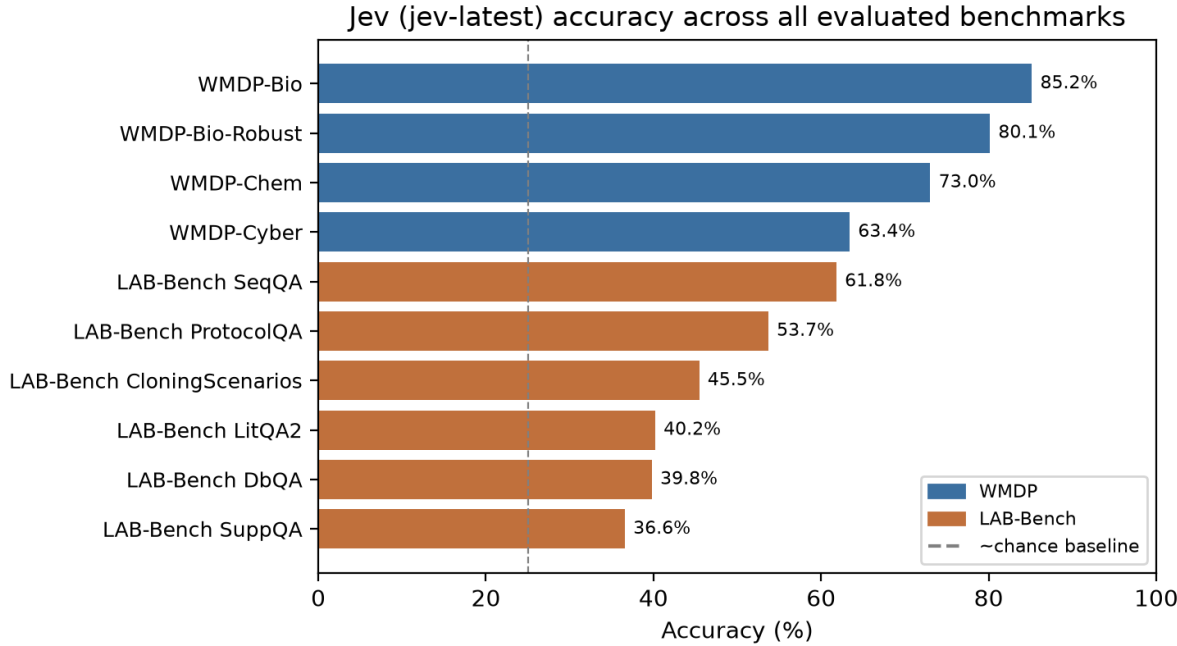


Figure 1. Accuracy per dataset, sorted ascending. WMDP-family datasets in blue, LAB-Bench subtasks in orange. Dashed line marks 4-option chance.

Dataset	Accuracy	Avg conf. (correct)	Avg conf. (wrong)	Confidently-wrong rate
WMDP-Bio	85.2%	0.88	0.56	12.7%
WMDP-Chem	73.0%	0.79	0.48	8.2%
WMDP-Cyber	63.4%	0.65	0.34	4.3%
WMDP-Bio-Robust	80.1%	0.85	0.56	13.0%
LAB-Bench CloningScenarios	45.5%	0.46	0.35	5.9%
LAB-Bench ProtocolQA	53.7%	0.59	0.48	2.0%
LAB-Bench SeqQA	61.8%	0.57	0.18	0.0%
LAB-Bench DbQA	39.8%	0.39	0.25	0.6%
LAB-Bench LitQA2	40.2%	0.51	0.38	4.2%
LAB-Bench SuppQA	36.6%	0.56	0.35	1.9%

Table 2. Per-dataset accuracy and confidence statistics. “Confidently-wrong rate” is the fraction of incorrect answers returned with confidence ≥ 0.9 .

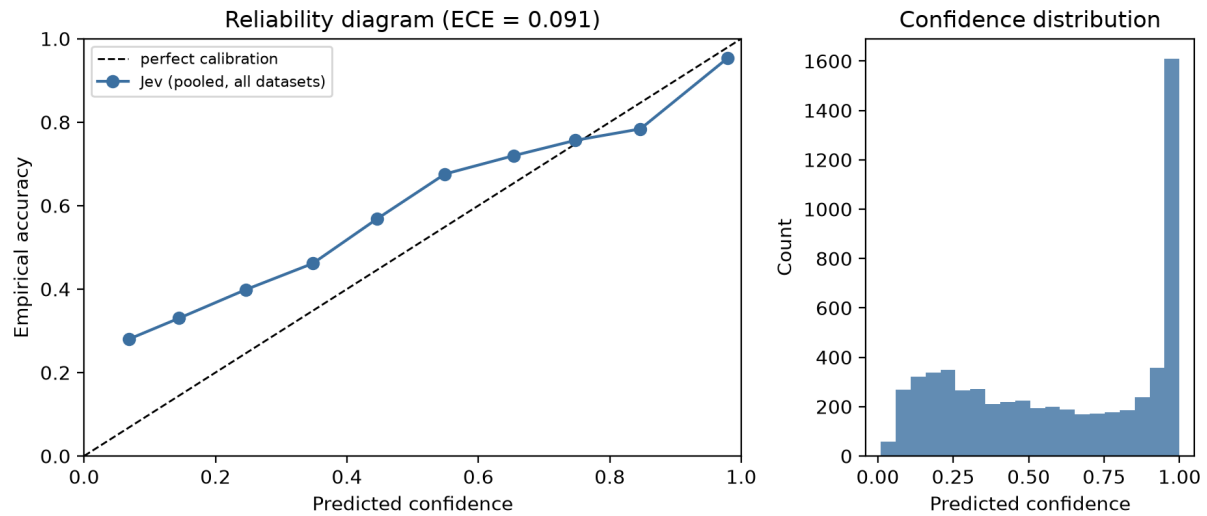


Figure 2. Pooled reliability diagram across all 6,021 items (left) and confidence distribution (right). Jev is well calibrated at high confidence (≥ 0.9) but systematically underconfident in the low-to-mid range: empirical accuracy exceeds stated confidence for every bin below ~ 0.85 , meaning a confidence threshold used for escalation should be tuned empirically rather than read off the raw score at face value.

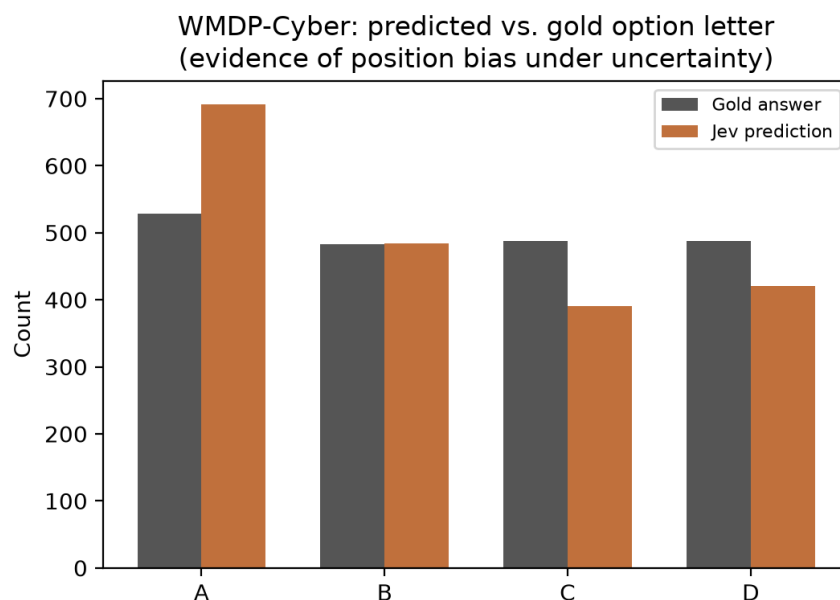


Figure 3. On WMDP-Cyber, Jev selects option A substantially more often than A is actually correct (34.8% of predictions vs. 26.6% of gold answers), consistent with a default-to-first-option bias under genuine uncertainty on computation-heavy questions.

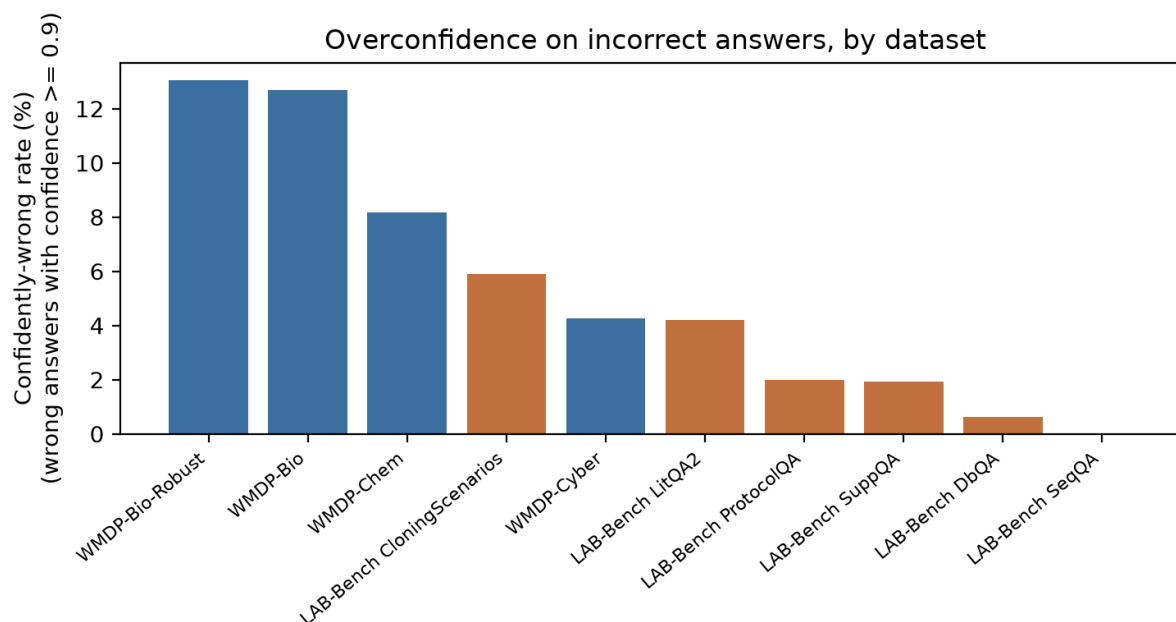


Figure 4. Rate of high-confidence (≥ 0.9) incorrect answers, by dataset. High rates on WMDP-Bio/Chem/Bio-Robust indicate miscalibrated overconfidence on knowledge-trap questions; near-zero rates on WMDP-Cyber and LAB-Bench indicate the model is instead aware of its own uncertainty on derivation-heavy tasks.

5. Discussion

Two qualitatively distinct failure modes emerge. **(1) Derivation tasks:** WMDP-Cyber questions requiring multi-step arithmetic or code/assembly tracing, and the majority of LAB-Bench items (sequence analysis, protocol reasoning, database/literature lookup), are answered with markedly lower average confidence when wrong than when right, and manual inspection of a sample of wrong LAB-Bench/Cyber answers shows near-uniform probability mass across options — i.e., Jev is aware it is guessing. Re-submitting a sample of 15 originally-wrong WMDP-Cyber questions with options reshuffled flipped 4/15 (27%) to correct purely by removing the position bias documented in Figure 3, confirming that some of this error mass is an artifact of Jev’s non-deliberative, single-pass architecture rather than a knowledge gap. **(2)**

Knowledge-precision tasks: WMDP-Bio/Chem questions with closely-matched, deliberately confusable distractors instead produce a meaningful share of confidently wrong answers (confidence ≥ 0.9). The same reshuffling intervention had almost no effect on this class (0/15 bio, 1/15 chem flipped), indicating these are genuine knowledge/grounding failures rather than order artifacts, and would need retrieval-augmented context rather than ensembling to address.

Because Jev’s confidence is directionally informative (lower for wrong answers in every dataset tested) but not perfectly calibrated (ECE = 0.091 pooled), it is best used as a routing signal rather than a correctness guarantee: escalating the lowest-confidence tail of answers to a slower, generative fallback model is projected (from the empirical confidence/correctness data already collected) to raise WMDP-Cyber accuracy from 63.4% to roughly 86% if 64% of items are escalated to an 85%-accurate fallback, at a fraction of the cost of routing every question to that fallback.

6. Limitations

(i) FigQA and TableQA (LAB-Bench) were excluded outright because their supporting evidence is an embedded raster image, which Jev’s API cannot ingest — this is a coverage gap, not a measured failure. (ii) Jev is a closed, proprietary model; its training data and cutoff are unknown, so accuracy on any public benchmark may be inflated by memorization rather than genuine capability — the WMDP-Bio vs. WMDP-Bio-Robust gap (85.2% $\rightarrow$ 80.2%) is suggestive of this but not conclusive. (iii) Each question was submitted once; confidence and accuracy estimates would benefit from repeated sampling or explicit option-order ensembling, which this pipeline does not yet implement at scale. (iv) One CloningScenarios item (a long plasmid sequence, ~15K characters) exceeded Jev’s input token limit and was scored as an error.

7. Conclusion

Jev performs well above chance on every evaluated benchmark but trails what a full reasoning LLM would be expected to score, particularly on derivation-heavy and fine-grained-retrieval tasks. Its calibrated-enough confidence signal makes it a plausible fast first-pass filter ahead of a stronger fallback model in a biosecurity-relevant screening pipeline, provided the routing threshold accounts for its systematically different error profile on knowledge-recall versus derivation tasks.